

Assignment 3 Solutions

Continuous Random Variables – Part 1

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Question 1 (5 marks)**Solution.**

Let

$$f_X(x) = \frac{\lambda}{2} e^{-\lambda|x|}, \quad \lambda > 0.$$

We solve each part step by step.

- (a) Verify that $f_X(x)$ satisfies the normalization condition.

For a valid PDF, we must have

$$\int_{-\infty}^{\infty} f_X(x) dx = 1.$$

Substituting the given PDF,

$$\int_{-\infty}^{\infty} \frac{\lambda}{2} e^{-\lambda|x|} dx.$$

Since

$$|x| = \begin{cases} -x, & x < 0, \\ x, & x \geq 0, \end{cases}$$

we split the integral into two parts:

$$= \frac{\lambda}{2} \left(\int_{-\infty}^0 e^{\lambda x} dx + \int_0^{\infty} e^{-\lambda x} dx \right).$$

Now evaluate each integral:

$$\int_{-\infty}^0 e^{\lambda x} dx = \left[\frac{e^{\lambda x}}{\lambda} \right]_{-\infty}^0 = \frac{1}{\lambda},$$

and

$$\int_0^{\infty} e^{-\lambda x} dx = \left[-\frac{e^{-\lambda x}}{\lambda} \right]_0^{\infty} = \frac{1}{\lambda}.$$

Therefore,

$$\int_{-\infty}^{\infty} f_X(x) dx = \frac{\lambda}{2} \left(\frac{1}{\lambda} + \frac{1}{\lambda} \right) = \frac{\lambda}{2} \cdot \frac{2}{\lambda} = 1.$$

Hence,

$$\boxed{\int_{-\infty}^{\infty} f_X(x) dx = 1}$$

so $f_X(x)$ is a valid probability density function.

(b) Find the mean of X .

By definition,

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx.$$

Thus,

$$E[X] = \int_{-\infty}^{\infty} x \cdot \frac{\lambda}{2} e^{-\lambda|x|} dx.$$

Notice that

$$x e^{-\lambda|x|}$$

is an odd function because x is odd and $e^{-\lambda|x|}$ is even.

Since the limits are symmetric about zero,

$$\int_{-\infty}^{\infty} (\text{odd function}) dx = 0.$$

Therefore,

$$\boxed{E[X] = 0}$$

This also follows from the fact that the distribution is symmetric about 0.

(c) Find the variance of X .

Recall,

$$\text{Var}(X) = E[X^2] - [E(X)]^2.$$

Since $E[X] = 0$,

$$\text{Var}(X) = E[X^2].$$

Now,

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx.$$

Substituting,

$$= \int_{-\infty}^{\infty} x^2 \cdot \frac{\lambda}{2} e^{-\lambda|x|} dx.$$

Since the integrand is even,

$$= \lambda \int_0^{\infty} x^2 e^{-\lambda x} dx.$$

Using the standard result

$$\int_0^{\infty} x^2 e^{-\lambda x} dx = \frac{2}{\lambda^3},$$

we get

$$E[X^2] = \lambda \cdot \frac{2}{\lambda^3} = \frac{2}{\lambda^2}.$$

Hence,

$$\boxed{\text{Var}(X) = \frac{2}{\lambda^2}}$$

□

Question 2 (4 marks)

Solution.

A point is chosen uniformly inside a triangle of height h .

Let X denote the distance of the point from the base of the triangle.

Since the point is chosen uniformly, probability is proportional to area.

- (a) Find the cumulative distribution function (CDF) of X .

By definition,

$$F_X(x) = P(X \leq x).$$

For $0 \leq x \leq h$, the event $X \leq x$ means the point lies in the lower portion of the triangle.

Instead of directly finding this area, it is easier to use the upper smaller triangle.

The height of the upper triangle is

$$h - x.$$

Since similar triangles have areas proportional to the square of their heights,

$$\frac{\text{Area of upper triangle}}{\text{Total area}} = \left(\frac{h - x}{h} \right)^2.$$

Therefore,

$$P(X > x) = \left(\frac{h - x}{h} \right)^2.$$

Hence,

$$F_X(x) = 1 - \left(\frac{h - x}{h} \right)^2, \quad 0 \leq x \leq h.$$

Thus,

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - \left(\frac{h - x}{h} \right)^2, & 0 \leq x \leq h, \\ 1, & x > h. \end{cases}$$

(b) Find the probability density function (PDF) of X .

Differentiate the CDF:

$$f_X(x) = \frac{d}{dx} F_X(x).$$

So,

$$f_X(x) = \frac{d}{dx} \left[1 - \left(\frac{h - x}{h} \right)^2 \right].$$

This gives

$$f_X(x) = \frac{2(h-x)}{h^2}, \quad 0 \leq x \leq h.$$

Therefore,

$$f_X(x) = \begin{cases} \frac{2(h-x)}{h^2}, & 0 \leq x \leq h, \\ 0, & \text{otherwise.} \end{cases}$$

□

Question 3 (4 marks)

Solution.

Calamity Jane goes to the bank and is equally likely to find:

0 customers ahead or 1 customer ahead.

If one customer is present, the service time is exponentially distributed with parameter λ .

- (a) Define a suitable random variable representing Jane's waiting time.

Let

W = Jane's waiting time.

Then:

- With probability $\frac{1}{2}$, there is no customer ahead, so

$$W = 0.$$

- With probability $\frac{1}{2}$, there is one customer ahead, and Jane must wait for that service time.

Let

$$T \sim \text{Exponential}(\lambda).$$

Then

$$W = T.$$

Thus,

$$W = \begin{cases} 0, & \text{with probability } \frac{1}{2}, \\ T, & \text{with probability } \frac{1}{2}. \end{cases}$$

(b) Derive the cumulative distribution function (CDF) of Jane's waiting time.

For $t \geq 0$,

$$F_W(t) = P(W \leq t).$$

Using total probability,

$$F_W(t) = P(W \leq t \mid W = 0) \cdot \frac{1}{2} + P(W \leq t \mid W = T) \cdot \frac{1}{2}.$$

Since $W = 0$ certainly satisfies $W \leq t$ for $t \geq 0$,

$$P(W \leq t \mid W = 0) = 1.$$

Also,

$$P(W \leq t \mid W = T) = P(T \leq t).$$

Since

$$T \sim \text{Exponential}(\lambda),$$

its CDF is

$$P(T \leq t) = 1 - e^{-\lambda t}.$$

Therefore,

$$F_W(t) = \frac{1}{2} + \frac{1}{2}(1 - e^{-\lambda t}).$$

Simplifying,

$$F_W(t) = 1 - \frac{1}{2}e^{-\lambda t}, \quad t \geq 0.$$

Hence,

$$F_W(t) = \begin{cases} 0, & t < 0, \\ 1 - \frac{1}{2}e^{-\lambda t}, & t \geq 0. \end{cases}$$

□

Question 4 (5 marks)**Solution.**

Consider a random variable X such that

$$X = \begin{cases} Y, & \text{with probability } p, \\ Z, & \text{with probability } 1 - p. \end{cases}$$

where Y and Z are continuous random variables with PDFs $f_Y(x)$ and $f_Z(x)$ respectively.

(a) Show that the PDF of X is

$$f_X(x) = pf_Y(x) + (1 - p)f_Z(x).$$

Let $F_X(x)$ be the CDF of X . Then

$$F_X(x) = P(X \leq x).$$

Using total probability,

$$F_X(x) = P(X \leq x \mid X = Y)P(X = Y) + P(X \leq x \mid X = Z)P(X = Z).$$

Hence,

$$F_X(x) = pP(Y \leq x) + (1 - p)P(Z \leq x).$$

That is,

$$F_X(x) = pF_Y(x) + (1 - p)F_Z(x).$$

Differentiating both sides with respect to x ,

$$f_X(x) = pf_Y(x) + (1 - p)f_Z(x).$$

Therefore,

$$\boxed{f_X(x) = pf_Y(x) + (1 - p)f_Z(x)}$$

(b) Consider the two-sided exponential distribution with PDF

$$f_X(x) = \begin{cases} p\lambda e^{\lambda x}, & x < 0, \\ (1 - p)\lambda e^{-\lambda x}, & x > 0. \end{cases}$$

Find the cumulative distribution function (CDF).

By definition,

$$F_X(x) = P(X \leq x).$$

Case 1: $x < 0$

$$F_X(x) = \int_{-\infty}^x p\lambda e^{\lambda t} dt.$$

Integrating,

$$= p \left[e^{\lambda t} \right]_{-\infty}^x = pe^{\lambda x}.$$

Thus,

$$F_X(x) = pe^{\lambda x}, \quad x < 0.$$

Case 2: $x \geq 0$

First include the probability on the negative side:

$$P(X < 0) = p.$$

Then

$$F_X(x) = p + \int_0^x (1-p)\lambda e^{-\lambda t} dt.$$

Integrating,

$$= p + (1-p) \left[1 - e^{-\lambda x} \right].$$

Therefore,

$$F_X(x) = 1 - (1-p)e^{-\lambda x}, \quad x \geq 0.$$

Hence,

$$F_X(x) = \begin{cases} pe^{\lambda x}, & x < 0, \\ 1 - (1-p)e^{-\lambda x}, & x \geq 0. \end{cases}$$

□

Question 5 (4 marks)**Solution.**

Given

$$f_X(x) = \begin{cases} ce^{-x}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the constant c .Since $f_X(x)$ is a PDF,

$$\int_{-\infty}^{\infty} f_X(x) dx = 1.$$

Thus,

$$\int_0^{\infty} ce^{-x} dx = 1.$$

$$c \int_0^{\infty} e^{-x} dx = 1.$$

$$c [-e^{-x}]_0^{\infty} = 1.$$

$$c(0 + 1) = 1.$$

Hence,

$$\boxed{c = 1}$$

(b) Find the cumulative distribution function $F_X(x)$.For $x \leq 0$,

$$F_X(x) = 0.$$

For $x > 0$,

$$F_X(x) = \int_0^x e^{-t} dt.$$

$$= [-e^{-t}]_0^x = 1 - e^{-x}.$$

Therefore,

$$F_X(x) = \begin{cases} 0, & x \leq 0, \\ 1 - e^{-x}, & x > 0. \end{cases}$$

(c) Compute $P(2 < X < 5)$.

$$\begin{aligned} P(2 < X < 5) &= F_X(5) - F_X(2). \\ &= (1 - e^{-5}) - (1 - e^{-2}) = e^{-2} - e^{-5}. \end{aligned}$$

Thus,

$$P(2 < X < 5) = e^{-2} - e^{-5}$$

(d) Find the expected value $E[X]$.

$$E[X] = \int_0^{\infty} x e^{-x} dx.$$

This is the mean of an exponential distribution with parameter 1, so

$$E[X] = 1$$

□

Question 6 (6 marks)

Solution.

Given the Rayleigh PDF

$$f_X(x) = \begin{cases} \frac{x}{\sigma^2} e^{-x^2/2\sigma^2}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

(a) Find $E[X]$.

By definition,

$$\begin{aligned} E[X] &= \int_0^{\infty} x \cdot \frac{x}{\sigma^2} e^{-x^2/2\sigma^2} dx. \\ &= \frac{1}{\sigma^2} \int_0^{\infty} x^2 e^{-x^2/2\sigma^2} dx. \end{aligned}$$

Using the standard Rayleigh result,

$$E[X] = \sigma \sqrt{\frac{\pi}{2}}$$

(b) Find the cumulative distribution function $F_X(x)$.

For $x < 0$,

$$F_X(x) = 0.$$

For $x \geq 0$,

$$F_X(x) = \int_0^x \frac{t}{\sigma^2} e^{-t^2/2\sigma^2} dt.$$

Let

$$u = \frac{t^2}{2\sigma^2}.$$

Then

$$du = \frac{t}{\sigma^2} dt.$$

Hence,

$$F_X(x) = \int_0^{x^2/2\sigma^2} e^{-u} du = 1 - e^{-x^2/2\sigma^2}.$$

Therefore,

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-x^2/2\sigma^2}, & x \geq 0. \end{cases}$$

(c) If $X \sim \text{Exponential}(1)$ and $Y = \sqrt{2\sigma^2 X}$, show that $Y \sim \text{Rayleigh}(\sigma)$.

Since

$$X \sim \text{Exponential}(1), \quad f_X(x) = e^{-x}, \quad x \geq 0.$$

Given

$$Y = \sqrt{2\sigma^2 X},$$

we write

$$X = \frac{Y^2}{2\sigma^2}.$$

Then

$$\frac{dX}{dY} = \frac{Y}{\sigma^2}.$$

Using transformation of variables,

$$f_Y(y) = f_X\left(\frac{y^2}{2\sigma^2}\right) \left|\frac{dX}{dY}\right|.$$

Thus,

$$f_Y(y) = e^{-y^2/2\sigma^2} \cdot \frac{y}{\sigma^2}.$$

Hence,

$$f_Y(y) = \frac{y}{\sigma^2} e^{-y^2/2\sigma^2}, \quad y \geq 0$$

which is exactly the Rayleigh distribution.

Therefore,

$$Y \sim \text{Rayleigh}(\sigma)$$

□

Question 9 (4 marks)

Solution.

Given

$$X \sim N(0, 1) \quad \text{and} \quad Y \sim N(1, 4).$$

Since $Y \sim N(1, 4)$, we have

$$\mu_Y = 1, \quad \sigma_Y = 2.$$

(a) Find $P(X \leq 1.5)$ and $P(X \leq -1)$.

Since X is a standard normal random variable,

$$P(X \leq 1.5) = \Phi(1.5).$$

From the standard normal table,

$$\Phi(1.5) = 0.9332.$$

Hence,

$$P(X \leq 1.5) = 0.9332$$

Similarly,

$$P(X \leq -1) = \Phi(-1).$$

From the standard normal table,

$$\Phi(-1) = 0.1587.$$

Therefore,

$$P(X \leq -1) = 0.1587$$

(b) Find the PDF of $\frac{Y-1}{2}$.

Let

$$Z = \frac{Y-1}{2}.$$

Since

$$Y \sim N(1, 4),$$

standardizing gives

$$Z \sim N(0, 1).$$

Therefore, the PDF of Z is the standard normal PDF:

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty$$

(c) Find $P(-1 \leq Y \leq 1)$.

Standardize using

$$Z = \frac{Y-1}{2}.$$

Then

$$P(-1 \leq Y \leq 1) = P\left(\frac{-1-1}{2} \leq Z \leq \frac{1-1}{2}\right).$$

Thus,

$$P(-1 \leq Y \leq 1) = P(-1 \leq Z \leq 0).$$

Using standard normal tables,

$$P(-1 \leq Z \leq 0) = 0.3413.$$

Hence,

$$P(-1 \leq Y \leq 1) = 0.3413$$

□

Question 10 (3 marks)

Solution.

Given

$$f_X(x) = \begin{cases} \frac{x^2 + 2}{3}, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find $E(X^n)$.

By definition,

$$E(X^n) = \int_{-\infty}^{\infty} x^n f_X(x) dx.$$

Since the support is $0 \leq x \leq 1$,

$$E(X^n) = \int_0^1 x^n \cdot \frac{x^2 + 2}{3} dx.$$

Simplifying,

$$E(X^n) = \frac{1}{3} \int_0^1 (x^{n+2} + 2x^n) dx.$$

Now integrate term by term:

$$= \frac{1}{3} \left[\frac{x^{n+3}}{n+3} + \frac{2x^{n+1}}{n+1} \right]_0^1.$$

Substituting the limits,

$$E(X^n) = \frac{1}{3} \left(\frac{1}{n+3} + \frac{2}{n+1} \right).$$

Therefore,

$$E(X^n) = \frac{1}{3} \left(\frac{1}{n+3} + \frac{2}{n+1} \right)$$

(b) Find $\text{Var}(X)$.

Recall,

$$\text{Var}(X) = E(X^2) - [E(X)]^2.$$

First, for $n = 1$:

$$E(X) = \frac{1}{3} \left(\frac{1}{4} + \frac{2}{2} \right) = \frac{1}{3} \left(\frac{1}{4} + 1 \right) = \frac{5}{12}.$$

Next, for $n = 2$:

$$E(X^2) = \frac{1}{3} \left(\frac{1}{5} + \frac{2}{3} \right) = \frac{1}{3} \left(\frac{3+10}{15} \right) = \frac{13}{45}.$$

Thus,

$$\begin{aligned} \text{Var}(X) &= \frac{13}{45} - \left(\frac{5}{12} \right)^2 \\ &= \frac{13}{45} - \frac{25}{144}. \end{aligned}$$

Taking LCM,

$$= \frac{208}{720} - \frac{125}{720} = \frac{83}{720}.$$

Hence,

$$\text{Var}(X) = \frac{83}{720}$$

□

Question 11 (3 marks)

Solution.

Given

$$X \sim \text{Uniform}(0, 1)$$

so that

$$f_X(x) = 1, \quad 0 \leq x \leq 1.$$

Also,

$$Y = e^{-X}.$$

Since $0 \leq X \leq 1$,

$$e^{-1} \leq Y \leq 1.$$

(a) Find the CDF of Y .

By definition,

$$F_Y(y) = P(Y \leq y).$$

Thus,

$$F_Y(y) = P(e^{-X} \leq y).$$

Taking logarithm,

$$e^{-X} \leq y \iff -X \leq \ln y \iff X \geq -\ln y.$$

Hence,

$$F_Y(y) = P(X \geq -\ln y).$$

Since $X \sim U(0, 1)$,

$$P(X \geq a) = 1 - a.$$

Therefore,

$$F_Y(y) = 1 - (-\ln y) = 1 + \ln y, \quad e^{-1} \leq y \leq 1.$$

So,

$$F_Y(y) = \begin{cases} 0, & y < e^{-1}, \\ 1 + \ln y, & e^{-1} \leq y \leq 1, \\ 1, & y > 1. \end{cases}$$

(b) Find the PDF of Y .

Differentiate the CDF:

$$f_Y(y) = \frac{d}{dy}(1 + \ln y).$$

Thus,

$$f_Y(y) = \frac{1}{y}, \quad e^{-1} \leq y \leq 1$$

and zero otherwise.

(c) Find $E(Y)$.

Since

$$Y = e^{-X},$$

we use

$$E(Y) = E(e^{-X}) = \int_0^1 e^{-x} dx.$$

Integrating,

$$= [-e^{-x}]_0^1 = (-e^{-1}) - (-1).$$

Hence,

$$E(Y) = 1 - e^{-1}$$

□